

6 | MAPPING THE SEMICONDUCTOR VALUE CHAIN

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Executive summary

The semiconductor value chain is highly complex, characterised by globally distributed production processes and numerous interlinked segments. Comprehensive understanding and effective risk monitoring and management of this value chain are crucial but require addressing significant data gaps. This paper provides evidence on dependencies and vulnerabilities in the semiconductor value chain to help inform policymakers in considering policies to strengthen its resilience. Working together with stakeholders to diversify supply sources and enhancing international collaboration as well as knowledge sharing can help reduce trade dependencies and manage potential supply chain disruptions.

More specifically, this paper provides a first mapping of the semiconductor value chain, focusing on two essential aspects: i) the structure of the semiconductor value chain and ii) its main vulnerabilities and bottlenecks. It identifies key segments, actors, and inputs, and offers a detailed analysis of how the different components and processes within the value chain interact, as part of the complex interdependencies that characterise the industry.

UN Comtrade data used in this paper show that the number of trade dependencies for semiconductors is particularly high, having more than doubled from 2012-14 to 2020-22. Dependencies for raw materials and manufacturing equipment are also high and have increased considerably in the same period. The supply of borates, silicon wafers and various inputs for wafer fabrication is highly concentrated, with several Asian producers playing key roles. Detailed analysis of bilateral trade flows shows that, for numerous products, many economies increasingly rely on the People's Republic of China (hereafter “China”), with limited alternative suppliers. This reliance creates dependencies not only for semiconductors themselves but also for several inputs along the value chain.

Building on OECD Inter-Country Input-Output data, this paper provides a more comprehensive analysis of the concentration of intermediate inputs flows, differentiating between final and intermediate goods (e.g. packaged and unpackaged semiconductors) and identifies whether the production is intended for domestic use or export. Data show that about half of the intermediate inputs used globally in the semiconductor industry are components sourced from within the industry itself, highlighting a high degree of intra-industry dependency. In parallel, approximately one-third of intermediate outputs are sold to other semiconductor firms for further processing. This dynamic highlights the layered and iterative nature of semiconductor production. In other words, the industry is deeply interconnected: firms rely on one another not just for components, but also for successive processing stages—sending parts back and forth as they are gradually assembled, refined, or integrated into final semiconductor products. Additionally, over half of all semiconductor intermediate outputs go to China, highlighting both the concentration of production and the vulnerabilities in the event of supply chain disruptions.

Enhancing the resilience of the semiconductor value chain by mitigating vulnerabilities is an important topic for policymakers. The OECD supports these efforts by analysing different segments of the value chain, by providing policymakers with data to help identify bottlenecks, monitor market imbalances and manage disruptions in the value chain, and by supporting international dialogue and exchange on best policy practices.

1 Introduction

During its inaugural year in 2023, the OECD Semiconductor Informal Exchange Network (hereafter the Network) discussed the importance of understanding and monitoring the semiconductor value chain, as well as managing value chain risks. The Network highlighted data challenges and affirmed the need for a comprehensive understanding of the semiconductor value chain in detail. To support this objective, the Network developed a taxonomy of semiconductor products and production facilities (OECD, 2024^[1]) as a first step, with the longer-term goal of identifying and measuring key upstream and downstream activities in the semiconductor value chain.¹

Mapping the semiconductor value chain provides critical insights into the interdependencies between production segments and the flow of inputs and outputs throughout the manufacturing process. This understanding is essential for addressing data gaps, identifying vulnerabilities, and informing policy decisions, in line with the objectives of the Network (OECD, 2023^[2]).

This paper explores two key aspects of the semiconductor value chain: The first examines its structure to better understand the different segments, linkages, inputs and outputs. A thorough understanding requires first identifying the inputs (and their sources) and outputs (and their destinations) at different segments of the semiconductor manufacturing process. This understanding also helps identify other industries beyond semiconductors where the resilience of the whole ecosystem, i.e. the network of firms, workforce, processes, and sectors involved in or dependent on semiconductor production, could be strengthened.

The second aspect focuses on identifying bottlenecks and vulnerabilities in the semiconductor value chain. A deeper understanding of these vulnerabilities is crucial for policymakers to manage risks and respond effectively to potential disruptions. The industry may also benefit from better information about vulnerabilities further upstream in the value chain (i.e., beyond the first tier of suppliers). This entails important data challenges. The analysis in this paper builds on the previous identification of key segments and their interdependencies, leveraging available data sources.

While detailed, granular data are available for upstream and front-end manufacturing, data availability is comparatively limited for back-end manufacturing, as well as for Research and Development (R&D) and Design. The analysis builds on highly disaggregated UN Comtrade data to quantify input-output relationships within the value chain, which are further complemented by OECD Inter-Country Input-Output (ICIO) data to derive key information such as intermediate consumption. There are nevertheless important data aggregation challenges that limit the analysis, notably for semiconductor inputs. Further work will investigate each segment in more detail.

The remainder of this paper is structured as follows. Section 2 identifies key nodes, inputs and actors of the semiconductor value chain. Section 3 describes the value chain using trade data and ICIO tables. Section 4 concludes.

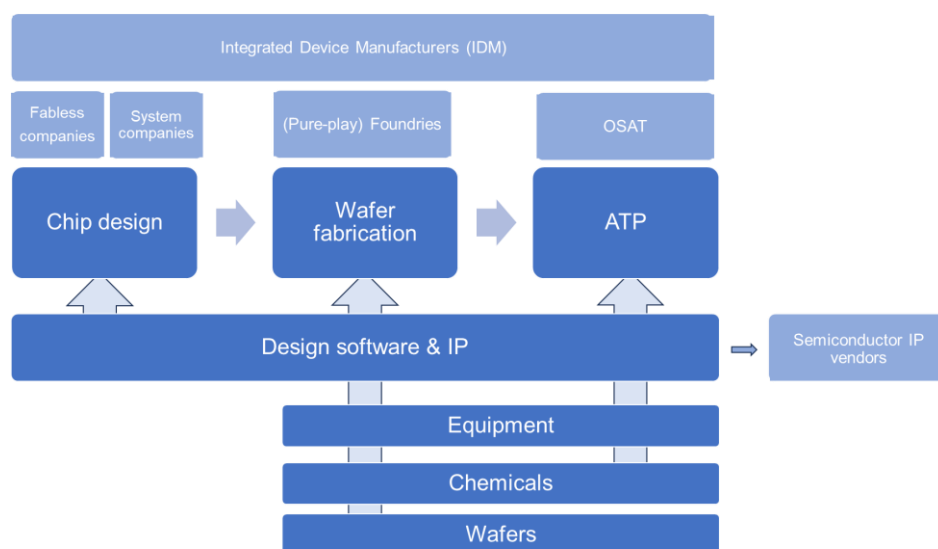
2 Overview of the semiconductor value chain

This section describes the semiconductor value chain in detail. For each key segment, the process is briefly explained, and the inputs (material and equipment) are outlined. A short list of key market players is provided for each.

Definitions and key segments

The semiconductor value chain is extremely complex and relies on a vast network of different types of actors with varying degrees of interaction. As shown in Figure 2.1, the semiconductor value chain can be segmented into three major process steps: design, wafer fabrication and assembly, test and packaging (ATP). Each of these process steps depends on distinct inputs, such as wafers (i.e. a thin slice of a semiconductor material, usually silicon, on which electronic circuits are created), equipment and materials, from often highly specialised suppliers. Both the process steps and their suppliers are heavily reliant on R&D, resulting in the semiconductor industry having among the highest R&D margins across industries (Bloom et al., 2020^[3]).

Figure 2.1. Key segments of the semiconductor value chain



Note: The light blue boxes indicate the different business models.

Each process step entails different structural characteristics, including barriers to entry. Chip design is skill- and research- intensive, while wafer fabrication is highly capital-intensive and relies on economies of scale. In contrast, ATP is less automated, making it more labour-intensive, with typically lower added value and

profit margins, compared to wafer fabrication and chip design. Because of these distinct differences between process steps, semiconductor companies have become more specialised, typically focusing on only one of the steps. This specialisation has contributed to the growing complexity and interdependence of the value chain over the last three decades. Below is a brief overview of the different business models that emerged and their relation to the three process steps.

- **Fabless companies** design and sell chips. They outsource front- and back-end manufacturing to foundries (i.e. manufacturing facilities producing chips for other companies) and Outsourced Semiconductor Assembly and Test (OSAT) companies, respectively. Fabless companies have among the highest R&D margins in the semiconductor industry, often spending 25% and more of their revenue on R&D (Kleinhans, 2021^[4]). Most of the fabless chip design companies are headquartered in the United States, such as Nvidia, Qualcomm and AMD.
- **System companies** design chips for their own products. They differ from fabless companies by selling systems (cars, smartphones, or services) rather than individual chips. Like fabless companies, they rely on foundries and OSATs for manufacturing. Examples of system companies with substantial chip design capabilities are Apple, Tesla, Google and Alibaba.
- **Semiconductor Intellectual Property (IP) vendors** design and sell functional blocks (also referred to as “IP cores”) that are used by chip designers to shorten time to market and lower chip design costs. Arm, Cadence, Synopsys and Imagination Technologies are among the largest IP vendors (Esteve, 2024^[5]).
- **(Pure-Play) Foundries** operate wafer fabrication plants (fabs) to provide contract wafer fabrication services to companies that design chips (fabless, Integrated Device Manufacturers (IDM) and system companies). They closely collaborate with IP vendors and Electronic Design Automation (EDA) suppliers (see below) to offer a “rich” IP value chain to their customers.
- **Integrated Device Manufacturers (IDM)** design, produce and sell chips. Traditionally, IDMs handled all three process steps – design, front- and back-end manufacturing – internally. Historically, this was the prevalent business model in the semiconductor industry. However, over the last two decades, IDMs have increasingly outsourced parts of their production, following the so-called “fab-lite” business model. Today, the “fab-lite” IDM model is prevalent, where IDMs decide on a product-by-product basis whether front- and/or back-end manufacturing is done in-house or outsourced. Some IDMs, such as Intel and Samsung, also offer foundry services for contract manufacturing. Unlike Pure-Play Foundries, **IDM-Foundries** use some or all of their fabs for contract manufacturing for external customers and internal production of its own chips.
- **Outsourced Semiconductor Assembly and Test (OSAT)** companies offer contract back-end manufacturing services to external customers.

It is important to understand that the level of vertical integration (across process steps) does not just vary at the company-level but already at the product level, illustrated by the following three examples — see (OECD, 2024^[1]) for an overview of product types and manufacturing process technologies.

Intel, like many other IDMs, has gradually shifted toward a fab-lite business model, outsourcing some elements of both front- and back-end manufacturing to foundries and OSATs (Mujtaba, 2024^[6]). At the same time, the company is expanding its foundry business to offer contract manufacturing to external customers for both front-end and back-end manufacturing. Thus, Intel could be considered a fab-lite IDM, an IDM-Foundry and an OSAT.

Infineon, Texas Instruments, Renesas and STMicroelectronics are examples of fab-lite IDMs in the analog, power, mixed-signal and microcontroller markets. While they may carry out front- and back-end manufacturing of a power semiconductor in-house, these same process steps could be entirely outsourced for microcontrollers. In the case of silicon-carbide (SiC) power semiconductors, some of these companies

are fully vertically integrated, from production of SiC ingots and epitaxy of the SiC wafer to packaging of the final chip (Yole Group, 2022^[7]).

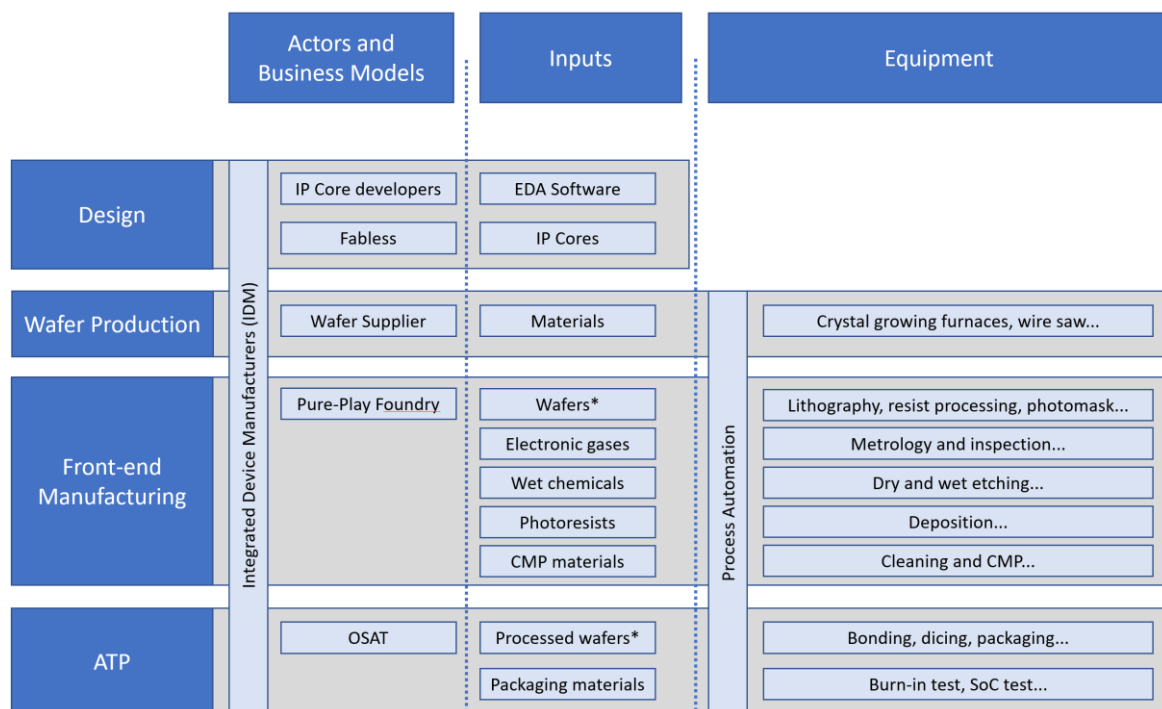
Alibaba, as a system company with their in-house chip design unit T-Head, is developing chips for the company's cloud business (Loghin, 2024^[8]). Additionally, Alibaba is developing IP cores and selling chips to customers (Chen et al., 2020^[9]). Thus, Alibaba's business could be characterised as a system company, a fabless company and an IP vendor.

These examples illustrate that semiconductor companies often do not have one fixed role in the value chain but engage in various activities along the value chain. The following subsection will provide a more detailed explanation of the different segments (process steps and their inputs) of the semiconductor value chain.

Actors and inputs across different segments

This section describes the semiconductor manufacturing process from design to ATP. Figure 2.2 shows how the key nodes of the value chain can be categorised into different groups of actors. An overview of the activities of each actor is provided along with an inventory of inputs and key market players therein.

Figure 2.2. Nodes in the semiconductor value chain



Note: * input from previous segment.

Source: OECD, based on CSET (2021^[10]) and ETO Supply Chain Explorer (2022^[11]).

Chip Design

Semiconductor development begins with digital chip design and an increasing number of companies are involved in chip design today. Regardless of whether fabless companies, system companies or IDMs designing a semiconductor, all rely on three key inputs: Electronic design automation (EDA) tools, IP cores and process design kits (PDK) (Ferguson, 2018^[12]).

- **IP cores** are pre-designed, reusable functional blocks used in chip designs. They are crucial to reducing development time, ensuring reliability, and meeting the stringent performance and power requirements of modern chips. Developing a high-quality IP block, such as a Central Processing Unit (CPU) core or a SerDes interface, requires years of expertise and domain-specific knowledge. Furthermore, (upfront) licensing fees and royalty structures (often paid on a per-unit basis) create ongoing financial and contractual dependencies between chip designer and IP vendor.
- **EDA tools** are specialised software used by engineers to bring together semiconductor designs using IP cores and custom designs. It allows them to design, simulate and verify the design. Modern chip design processes involve more than 40 individual steps and most chip designers “mix and match” EDA tools to select the best-in-class tool per step (Nenni, 2022^[13]). The costs of designing a chip vary widely depending on the feature size of the process node, the complexity of the design as well as license costs for EDA tools and IP cores (Shilov, 2023^[14]).
- **PDKs** ensure that semiconductor designs align with the capabilities of the foundry where the chip will be taped out. IDMs and foundries develop PDKs for their process nodes that contain all the design rules of a node, such as minimum distances between functional elements. While the IDM uses the PDK only internally, foundries provide the PDKs to their customers. The EDA software then uses the PDK’s design rules to ensure that the design created can be manufactured on this process node. Foundries regularly improve their PDKs and there is often close collaboration between EDA tool vendors, such as Cadence and Synopsys, and foundries.

Innovation in semiconductor design starts through novel designs of chips and IP cores. R&D is particularly crucial for optimising and overcoming the design challenges of creating the most efficient semiconductors within the current physical limitations of the fabrication process, while considering the planned end use (logic, memory, etc.).

R&D plays a critical role in all steps of semiconductors production. Innovations in electronics, material science, and miniaturisation, among others, can significantly impact the entire value chain. This can, for example, manifest in improved designs, manufacturing processes or equipment for both state-of-the-art and next-generation semiconductor technologies. The fabrication process itself is an important area of ongoing R&D in companies and across the value chain (e.g., through institutions such as Interuniversity Microelectronics Centre in Leuven, Belgium). Therefore, many key technology partners and input providers throughout the semiconductor value chain invest heavily in R&D.

Landscape

The current landscape of companies designing chips is quite diverse. Traditionally IDMs, such as NXP, Renesas and Texas Instruments, as well as fabless companies, such as AMD, Broadcom and MediaTek, were designing chips. While this remains the case, system companies, such as Alibaba, Apple and Google, increasingly invest in their internal chip design (Ma, 2022^[15]). As a result, it is challenging to map the global chip design landscape and compare chip design capacities of different economies solely based on data from IDMs and fabless companies.

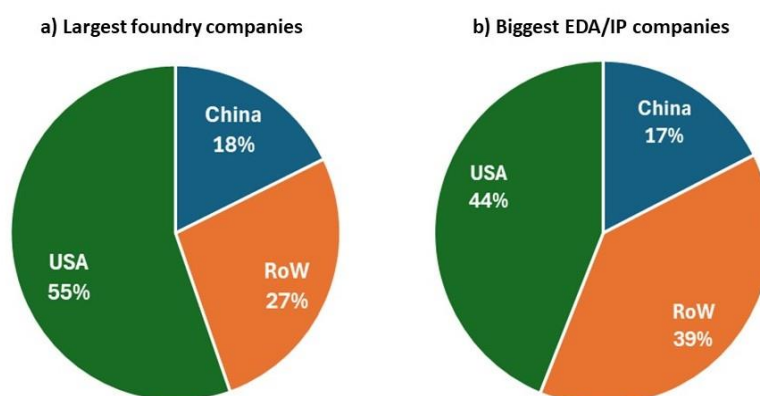
One way to better understand chip design capacity of different economies is to examine revenue shares by geography from the largest foundries. This approach captures all design activities from fabless and system companies, as well as some activities from fab-lite IDMs, which also outsource parts of their manufacturing.

Figure 2.3, left panel, shows the revenue shares by geographical region from TSMC, Globalfoundries, UMC, SMIC and Hua Hong in 2023. Together, these five foundries account for 80% of the global foundry market. Customers headquartered in the United States accounted for roughly 55% of sales, while customers in the People’s Republic of China (hereafter “China”) generated the second-highest sales, at 18%. The high share of foundry customers headquartered in the United States is not surprising, given that

many of the largest fabless and system companies, such as AMD, Apple, Nvidia and Qualcomm, are headquartered there (BCG, SIA, 2022^[16]).

Figure 2.3. Annual revenue shares in 2023 from different geographical regions for selected companies

Panel a: % of foundry revenue by geography; panel b: % of EDA/IP revenue by geography



Note: The left panel includes the following foundry companies: TSMC, UMC, SMIC, GlobalFoundries, and Hua Hong Semiconductors which make up approximately 80% of the market share. The value for China and the United States may be underreported due to geographical region categorisation by each of the companies. A further breakdown of “Rest of the World” is not possible given that regional categories widely differ between these five foundries. The right panel includes the following EDA/IP companies: Cadence, Synopsys Inc., and Arm.

Source: OECD based on TSMC (2024^[17]), UMC (2023^[18]), SMIC (2024^[19]), GlobalFoundries Inc. (2023^[20]), and Hua Hong Semiconductor Limited (2023^[21]) (left panel) and Cadence Design Systems (2023^[22]), Synopsys Inc (2023^[23]), and Arm Holdings plc (2023^[24]) (right panel)

Another perspective is to examine the geographical revenue distribution of EDA and IP vendors. Since both markets are highly concentrated, looking at geographical revenue shares of the largest players provides insight into which economies spend the most on EDA and IP, indicating the most commercially active chip design value chain. Figure 2.3, right panel, shows the revenue share by geographical region for Cadence, Synopsys and Arm. Together, these companies account for more than 60% of the global EDA market and 70% of the IP market (Zeoli, 2024^[25]; Esteve, 2024^[5]). This figure closely resembles the left panel, with the notable difference that the share of the United States is lower, at 44% instead of 55%. One explanation is that other economies, such as Japan, Korea and the European Union, are largely dominated by IDMs, which also license EDA software and IP from these vendors.

Tools and equipment

- **EDA Tools:** the market is concentrated in the United States through three companies, namely Cadence, Siemens EDA (formerly Mentor Graphics), and Synopsys (Kleinhans, 2022^[26]).
- **Core IP:** the market is split between the United States and the United Kingdom through companies such as Arm, Synopsys, Cadence, SST and Imagination Technologies (Clarke, 2024^[27]).

Wafer production

Wafers can consist of various materials, such as silicon, silicon carbide (SiC), gallium nitride and diamond. However, silicon is by far the dominant wafer material and the backbone of today's semiconductor manufacturing, especially for logic and memory chips. In 2023, more than 250 million (8-inch equivalent) silicon wafers were shipped (SEMI, 2024^[28]), compared to an estimated 1.7 million SiC wafers (irrespective of diameter) (Huang, 2023^[29]). The production process for different types (materials) of wafers varies, but

the goal remains the same: to achieve high purity, uniformity, and in the long-term an ever-increasing wafer diameter (mks, 2024^[30]).

Fabs typically obtain raw wafers from specialised providers. In order to deliver a raw wafer, the specialised production facility has to produce a cylinder (referred to as a boule or ingot) of pure silicon (or another material). Wafer thin discs are then cut from the cylinder with the help of special machine tools. The level of semiconductivity can then be altered across different parts of the wafer by using ion implanters to embed “dopants”. The latter step can also be performed in the fab as part of chips fabrication. Key inputs to this step include raw semiconducting materials, typically silicon, which are usually refined in a separate facility, as well as furnaces and machine tools for forming and slicing the ingot, implanters and dopant substances for treating the wafers.

Landscape

More than 50% of the global silicon wafer market are supplied by two Japanese companies, Shin-Etsu and SUMCO, with other significant players including GlobalWafers from Chinese Taipei and Siltronic from Germany (Siltronic, 2024^[31]). However, the landscape for other types of wafers, such as silicon-carbide, gallium nitride, sapphire or diamond wafers is fundamentally different. While silicon wafers will continue to dominate, especially for logic and memory chip production, alternative wafer materials provide superior physical characteristics for power and radio frequency chips. The production process and corresponding supply chain for these wafers substantially differs from that of silicon wafers.

Material inputs

Polycrystalline silicon, or polysilicon, is ultra refined metallurgical grade silicon created from quartz. The quartz undergoes multiple purification steps to remove as many impurities as possible, resulting in silicon rods. These rods are key inputs for wafer fabrication, as they allow for the creation of an ingot which will eventually be machined into wafers. The process is very similar for germanium wafers.

Other semiconductor materials can be used as substrates, including, but not limited to, silicon carbide, gallium arsenide, aluminium gallium phosphide, and mercury cadmium telluride. Each requires the eponymous elements in high purity. Nevertheless, the vast majority of semiconductor production currently uses silicon wafers. Although the production of most of these raw materials is largely concentrated in China, many of these elements are present across the globe.

Tools and equipment

- **Crystal growing furnaces:** the market is concentrated in a German company called PVA TePla.
- **Crystal machining tools:** the market is concentrated in Japanese companies Accretech, Okamoto, and Disco.
- **Ion implanters:** the market is concentrated in the United States with one Japanese competitor. Main companies include Applied Materials (United States), Nissin Ion (United States), SMIT (Japan), and Axcelis (United States).

Source: (Center for Security and Emerging Technology, 2021^[10])

Fabrication (front-end manufacturing)

Semiconductor fabrication facilities (or fabs) take raw wafers and create integrated circuits (chips) on them through a multistep process. This typically involves a cycle of polishing, deposition, resist application, lithography, etching, ion implantation, and resist removal, repeated according to the number of layers required for the final chip (Halbleiter, 2024^[32]). This description is vastly simplified and does not reflect the

complexity and sensitivity of the process. Indeed, the most advanced fabrication facilities are some of the most complex and expensive production facilities, taking years to build. See the OECD semiconductor taxonomy (OECD, 2024^[1]) for further information on semiconductor types and production facilities.

While semiconductor fabrication itself is important, it relies on a wide array of inputs both in equipment and materials. Many of these inputs are specific to semiconductor fabrication and are highly specialised. A prime example are Extreme Ultraviolet (EUV) lithography machines produced by the Dutch company ASML. Modern semiconductor production depends on various lithography tools, such as wet and dry deep ultraviolet lithography tools. For the most advanced production processes at 5nm and below, several layers of a chip can only be produced with EUV equipment. Nonetheless, “older” deep ultraviolet (DUV) lithography equipment is still required for the other layers (Kleinhans and Lee, 2024^[33]). Furthermore, for photomasks production and advanced packaging, other types of lithography equipment, such as X-ray lithography and electron-beam lithography, are essential.

This step of the semiconductor manufacturing process requires the most inputs in terms of equipment and materials, with wafers requiring different specialised tools and inputs as they are processed. Throughout its journey in the fab, a wafer is constantly moved and monitored, requiring specific process controlling and wafer handling equipment.

Landscape

It is important to understand that wafer fabrication is highly diversified, and market analyses need to consider differences in node density, process technology and business models, at the very least. For instance, a modern power semiconductor cannot be built on a process node for high-performance processors or on one for memory chips. Furthermore, some chip types are mainly built in an IDM or a foundry business model. Commodity memory, such as DRAM and NAND, is solely built by IDMs, including Samsung, SK Hynix, Micron, Kioxia and YMTC. These IDMs are also among the companies with the highest wafer capacity, as competition in commodity memory is driven by economies of scale. In contrast, most of today’s smartphone System on Chip (SoC) and Artificial Intelligence (AI) accelerators are designed by fabless and system companies and then manufactured by foundries.

Material inputs

Photomasks: The material and functioning of the photomask depend on the type (or generation) of lithography tool. They allow transferring the pattern onto chips using photolithography. While DUV lithography tools use transmissive masks, often made of glass substrates which are coated with opaque materials (typically chromium), EUV lithography uses reflective masks consisting of multiple layers. Photomask production is either done by the fab internally or by merchant mask shops. It starts with a mask blank that is then processed to create the pattern of each layer of the chip. According to the Center for Security and Emerging Technology (2021^[10]), the production of transmissive masks (DUV lithography) is not particularly concentrated, and the raw material components are only moderately concentrated, even if chromium exports are concentrated, South Africa being the largest exporter (US Geological Survey, 2024^[34]).

Photoresists: these are chemicals that are deposited on the wafers that undergo a chemical reaction when exposed to the light passing through the photomask. This reaction lays the groundwork for etching, which is performed on areas where the photoresist has dissolved. Each generation of lithography uses photoresists that are specifically tuned to the wavelength of the light source. The shorter the wavelength, the more advanced and specialised the resist chemistry must be to meet the increased demands of finer feature sizes. Photoresists are organic materials typically made up of resin, sensitiser, and solvent. Japan is the dominant player in the photoresist market (Center for Security and Emerging Technology, 2021^[10]).

Wet processing chemicals: these chemicals are used for different purposes and process steps. For example, wet processing chemicals for etching are used after lithography to transform solid material into liquid to etch the chip layout. Wet processing chemicals are also used during the various cleaning steps to remove residues. Examples of wet processing chemicals are sulfuric acid, isopropyl alcohol, ammonium hydroxide, phosphoric acid, nitric acid, hydrochloric acid, and acetic acid. Some of these require ultra-high levels of purity and therefore are only produced by a limited number of plants.

Specialty gases: gases are used throughout the semiconductor fabrication process as dopants, dry etchants, in the dry etching process or for protection from atmospheric exposure. These include common gases like nitrogen, hydrogen, and carbon dioxide, electronics specialty gases, such as nitrogen trifluoride, tungsten hexafluoride, hydrogen chloride, ammonia, disilane, germane, high purity carbon dioxide, and nitrous oxide; or noble gases including neon, argon, helium and krypton. The level of concentration of production of these gases varies. Some are widely available while others are produced by only a few specialised companies (SIA, 2021^[35]).

Chemical mechanical planarisation (CMP) slurries: a uniform surface for lithography requires wafers to be planarised using a polishing pad and a slurry, which contains abrasives and chemicals. These include rare earths such as cerium. The production of these slurries and pads is concentrated in Japan (Center for Security and Emerging Technology, 2021^[10]).

Sputtering targets: these are used for Physical Vapor Deposition (PVD) to apply a thin film through the sputtering method which consists of ejecting material from the target on to the wafer. The most commonly used sputtering targets in semiconductor manufacturing are made of aluminium, titanium, tantalum and copper. The choice of target material depends on the type of chip and desired properties. Japan and the United States are the main producers of targets for the industry (Stanford Advanced Materials, 2023^[36]).

Tools and equipment

A large variety of chipmaking equipment is needed in modern wafer fabrication. In addition to equipment for the major process steps – lithography, etching, deposition –, equipment is needed for cleaning, CMP, process automation and metrology and inspection. To understand the landscape, competitive positions and dependencies, these broad categories need to be further broken down. For example, wafer fabrication depends on several different types of deposition equipment in different process steps but also on the type of chip that is built. While Applied Materials is leading the market for Physical Vapor Deposition (PVD) equipment (Castellano, 2023^[37]), ASM and Tokyo Electron are the market leaders in Atomic Layer Deposition (ALD) (Fund, 2024^[38]). A similar situation exists in other chipmaking equipment categories, such as etching, bonding and cleaning equipment, where technology leadership depends on the specific subcategory. Thus, the following list of equipment suppliers should only be seen as a broad overview.

- **Advanced photolithography equipment:** the market is primarily dominated by companies from the Netherlands and Japan, including ASML (Netherlands), Canon (Japan), and Nikon (Japan). ASML is the only firm that provides EUV equipment. Notably, equipment suppliers also depend on various highly specialised component and sub-system suppliers. For instance, ASML depends on over five thousand suppliers and builds only around 15% of a lithography machine itself (Kleinhans and Lee, 2024^[33]).
- **Advanced photomasks:** mask blanks are mainly supplied by Japanese companies, such as Shin-Etsu, Hoya and AGC. The production of photomasks by merchant mask shops is mainly led by companies from Japan and the United States, such as Dai Nippon and Photronics respectively.
- **Deposition tools:** the market is shared between the United States (Applied Materials, Lam Research), Japan (Tokyo Electron), and the Netherlands (ASM).
- **Etching tools:** dry etching (using gases or plasma in a vacuum to remove material) and wet etching (using liquid chemicals) are the two general types of etching equipment used in wafer

fabrication. While wet etching tools are mainly used for cleaning and residual film removal, dry etching is indispensable to build the structures of a chip. There are several different dry etching technologies and different companies are leading in the respective technologies (Huff, 2021^[39]). Lam Research (United States), Applied Materials (United States) and Tokyo Electron (Japan) are among the leading companies in the dry etch tool market (Tsutsumi, 2024^[40]).

- **Wafer cleaning tools:** Screen (Japan), Tokyo Electron (Japan) and Lam Research (United States) accounted for 75% of the wafer cleaning equipment market in 2022 (Hars, 2022^[41]). Since 2020, Chinese companies, such as ACM Research and Naura also gained market shares (Wen, McKay and Ma, 2022^[42]).
- **Resist processing tools:** used to coat, develop and bake the photoresist on the wafer before and after the lithography process step. Leading suppliers are Japanese companies Tokyo Electron and Screen.
- **Wafer bonders and aligners:** joining two or more wafers either permanently, to form stacked structures, or temporarily, where a support wafer is attached to aid in processes like thinning or alignment before being de-bonded. Leading companies are EV Group (Austria), SÜSS MicroTec (Germany), Tokyo Electron (Japan) and Applied Materials (United States).
- **Automated material handling systems:**
The market for photomask handlers is split between the United States and Japan through Brooks and Daifuku, respectively. Wafer handlers are mainly supplied by the Japanese companies Daifuku and Rorze.
- **Process monitoring:** main companies include Keysight Tech, Keithley Instruments in the United States.

Source: (Center for Security and Emerging Technology, 2021^[10])

Assembly, test and packaging (back-end manufacturing)

In this step, the processed wafers are cut into individual dies, i.e. the bare (silicon) substrate that contains the integrated circuit. These dies are then packaged individually into protective frames surrounded by a resin shell. The frame allows the final chips to be soldered onto printed circuit boards (PCB). Throughout this process the chips are rigorously tested to ensure that there are no defects and that they are functioning properly. The chips can be mounted onto a PCB which is shipped to a different assembly facility while in some cases the circuit board is assembled into the final product in the same facility. Collectively these process steps are called ATP or back-end manufacturing.

There is an ever-increasing variety of packaging technologies, and chip designers typically choose a specific packaging technology based on performance requirements, cost constraints, form factor and reliability. The most common categories of packaging technologies are wire bonding, flip chip, wafer-level packaging, system-in-package (SiP) and 3D packaging (Burkacky, Kim and Yeom, 2023^[43]). Packaging technologies are steadily evolving and IDMs, foundries and OSATs offer increasingly specialised packaging approaches targeted for specific applications, such as high-performance computing, automotive and mobile chips (Vardaman, 2021^[44]). For example, advanced packaging techniques such as 2.5D and 3D packaging are increasingly popular, particularly for mobile applications and high-performance computing. These techniques involve stacking multiple dies or integrating different functions onto a single package, improving performance and reducing the space needed (VerWey, 2022^[45]).

Advanced packaging also makes mapping the ATP landscape increasingly complex since Pure-Play foundries, IDM-foundries and (traditional) OSATs now offer advanced packaging processes. The reason is that advanced packaging depends on more sophisticated equipment, better cleanrooms and in general a more controlled environment, blurring the line between front- and back-end manufacturing (Patel, 2022^[46]).

The conventional ATP process requires many distinct tools and equipment to process the finished wafer. In terms of inputs, the assembly and packaging steps usually require more mechanical equipment and material inputs while the testing steps require specialised electronic equipment.

Landscape

In ATP, the segment is split between OSATs offering contract back-end manufacturing and IDMs that do ATP internally. However, as mentioned before, since most IDMs today follow a fab-lite business model, they decide on a product-by-product basis whether to outsource packaging to an OSAT or foundry.

Global revenues from OSAT companies reached USD 44 billion in 2022, with the largest OSAT companies (by revenue) being ASE Group (Chinese Taipei), Amkor (United States), JCET (China) and Tongfu Microelectronics (China) (IDC, 2023^[47]). When focusing specifically on advanced packaging, Intel, TSMC and Samsung are also among the leading players (Yole Group, 2023^[48]).

Material inputs

The most critical inputs are finished wafers from fabs (IDMs or foundries). Traditional packaging materials tend to have less stringent specifications, making them more widely available due to the lower technological challenges of production. Given the concentration of ATP activities in Southeast Asia, the production of these packaging components is also somewhat concentrated in the region, particularly in Japan.

Lead frames: are metal structures that support and secure chips while carrying signals from the chip to the wiring board. They are made from thin metal plates that have been etched or stamped to desired patterns. They are usually made from copper, but alternative materials are copper alloys and iron-nickel alloys, which can be gold or silver plated.

Packaging substrates: chips must be protected and are therefore encased in substrates. The choice of substrates depends on the type of usage for the chip and the desired properties. The main types of packaging substrate materials are plastic, ceramic, metal, organic, and composites (Cadence Design Systems, n.d.^[49]). Advanced packaging substrates are on the cutting edge of development, including glass substrates for heterogeneous integration.

Bonding wires: connect the chip to the packaging and, in some cases, to other electronics. The main materials for these wires are copper, aluminium, gold, and silver.

Die-attach materials: are used to attach the chip (die) to a substrate or package (lead frame, ceramic substrates, PCBs, etc.). Their core functions are mechanical adhesion, electrical insulation, and thermal management. The three main categories of materials used for die-attaching are epoxy resin, metal alloys and ceramics (Ayodele, 2023^[50]).

Tools and equipment

- **Dicing tools:** concentrated in Japan through Disco and Accretech.
- **Integrated assembly tools:** these integrate multiple packaging steps and systems. Among the leading companies are ASM Pacific Technologies (ASMPT) (Singapore) and Besi (Netherlands).
- **Testing equipment:** split between Japan (Advantest), the United States (Teradyne) and Korea (UniTest).

Source: (Center for Security and Emerging Technology, 2021^[10])